

# How AI-generated images affect human preference and trust in websites

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**Figure 1: Reference images used for prompt creation: human (left), environment (middle), and objects (right). These served as the basis for generating consistent text-to-image prompts.**

## Abstract

AI-generated images (AIGIs) are increasingly integrated into websites, yet their effects on user perception in contexts where trust matters remain underexplored. We report a within-subjects online experiment (N=43) in which participants evaluated six websites containing either photographic imagery or AIGIs, first without and then with disclosure of AI use. Without disclosure, participants tended to prefer websites using AIGIs, while trust judgments were more evenly distributed. When AI use was disclosed, trust shifted clearly toward websites using photographic images. These effects varied by website context, with higher tolerance in entertainment settings and stronger trust penalties in government and health settings. Open responses indicate a tension between visual appeal and perceived authenticity. While polished AI imagery enhanced aesthetic appeal, disclosure raised concerns about authenticity and credibility. We show that disclosure reshapes trust in website contexts and that acceptance of AIGIs depends on contextual expectations rather than image quality alone.

## CCS Concepts

• Human-centered computing → Empirical studies in HCI.

## Keywords

AI-generated images, Generative AI, Website trust, User perception, Human-computer interaction

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## 1 Introduction

Generative AI has rapidly lowered the cost and effort of producing high-quality imagery for digital products and websites [5]. Contemporary models can produce photorealistic outputs that are difficult to distinguish from photographs [3, 9], and some work suggests that AI-created visuals are often rated highly on aesthetic dimensions [8]. As a result, AIGIs are increasingly used in web design, where images play an important role in shaping perceptions of professionalism, credibility, and organisational identity rather than merely serving a decorative function [6, 13].

HCI research has long shown that first impressions of websites form quickly and can shape later judgments [11, 14, 15]. Importantly, the standards users apply to a website depend on its category, such as entertainment versus government [16]. This distinction is particularly relevant for AIGIs. Prior work suggests that users apply different evaluative standards depending on website context and purpose [6, 16]. In contrast to advertising or social media, where synthetic or stylised visuals are often expected, organisational websites may be subject to stronger expectations of authenticity. These



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expectations are likely to be especially salient in domains such as government and healthcare, where credibility and trust play a central role.

Disclosure further complicates the adoption of AI-generated imagery. Across domains, transparency about AI creation has been shown to reduce perceived legitimacy and trust [7, 17], and brands that use AI-generated content are often perceived as less authentic [2]. However, much of the existing research examines AI-generated images either in isolation or within media contexts such as advertising and social media [2, 7, 8, 10]. Far less is known about how such images are perceived when embedded in organisational websites, where visual choices interact with established credibility expectations.

In this work, we examine how AIGIs influence preference and trust in website contexts and how disclosure relates to these judgments across different types of websites. Our findings indicate that disclosure is associated with a marked shift in trust toward websites using photographic imagery, even when AI-generated images are initially preferred. We further observe that preference and trust do not respond uniformly to disclosure and that tolerance for AI-generated imagery varies across website contexts, with stronger negative reactions in government and healthcare settings than in entertainment contexts. Finally, our qualitative findings point to a recurring tension between aesthetic appeal and perceived authenticity, suggesting that technical image quality alone may be insufficient to ensure acceptance in contexts where credibility is important.

## 2 Related Work

Recent advances in generative AI have enabled text-to-image models to produce visually appealing and increasingly photorealistic images [3, 5]. Prior work shows that preferences for AI-generated images are shaped by perceptual attributes such as lighting, saturation, and composition [8]. At the same time, typical users often struggle to distinguish AI-generated images from photographs [9], and visual realism has been shown to influence perceived trustworthiness [1, 4, 12]. Together, these findings suggest that high visual quality may positively shape first impressions, even when the synthetic nature of the content is not readily apparent.

Beyond perception, trust plays a central role in how AI-generated content is evaluated. Trust judgments are closely tied to perceptions of legitimacy, competence, and transparency [18]. Across domains, disclosure of AI involvement has been associated with reduced trust and more negative attitudes toward content and its creators [7, 17]. Related work in social media and branding further indicates that the use of AI-generated imagery can diminish perceived authenticity [2]. These findings highlight a recurring tension between the benefits of AI-generated visuals and the potential costs of transparency, motivating closer examination of disclosure effects in contexts where credibility matters.

Websites are a particularly relevant setting for studying these issues. Prior HCI research shows that design quality and professionalism shape trust judgments, and that user evaluations vary by website category and purpose [6, 14–16]. This suggests that images may be judged against different standards depending on context. While AI-generated images have been examined in isolation as well

as in advertising and social media contexts [2, 7, 8, 10], their use within organisational websites remains less explored, despite their role in establishing credibility and trust.

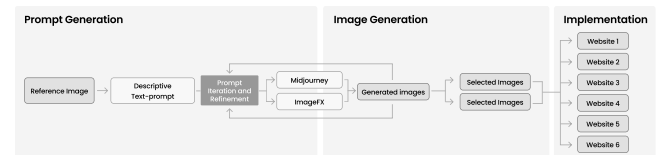
## 3 Study: AIGIs in Website Contexts

We conducted a within-subjects online experiment ( $N=43$ ). Participants evaluated six websites across two consecutive rounds. In the first round, participants compared two versions of each website, one using photographic imagery and one using AI-generated imagery, without being told which version was which. In the second round, participants re-evaluated the same websites after being informed which version used AI-generated images.

For each website in each round, participants made two forced-choice judgments. They indicated which version they preferred and which version they trusted more. The order of websites was randomised, and the position of the original and AI-generated versions was counterbalanced to reduce ordering effects. This design allowed us to examine both initial, unmediated impressions and shifts in preference and trust following disclosure.

### 3.1 Materials

We selected six websites, two per category, to represent contexts with different authenticity expectations: government and health, entertainment, and services. The websites were chosen to be non-Scandinavian and unlikely to be strongly familiar to Norwegian participants. Each site contained four to five images, including a prominent header image. For each website, the original images were replaced with matched AI-generated images produced using Midjourney and ImageFX, which were selected during stimulus preparation. Prompts were iteratively refined to match the content, composition, and tone of the original images. Example original website imagery is shown in Figure 1, and the selected websites are listed in the appendix (Table 1). Figure 2 illustrates the workflow used to generate and integrate the AI-generated images into the website stimuli. Stimulus preparation involved author judgment and was not intended as a systematic evaluation of model performance.



**Figure 2: Workflow of the image generation process for creating website stimuli.**

### 3.2 Participants

A total of 43 participants took part in the study (25 male, 18 female). Most were between 18 and 29 years of age and they had diverse educational and professional backgrounds spanning engineering and technology, design and media, and other fields. With respect to prior experience with artificial intelligence, most participants reported moderate to high familiarity with AI tools, while a smaller portion reported little or no prior experience.

### 3.3 Procedure

The study was conducted as an online survey and consisted of four parts. After providing informed consent and demographic information, participants completed two evaluation rounds followed by a short questionnaire. In the first round, participants viewed pairs of website versions, one using photographic imagery and one using AI-generated imagery, without being informed about the origin of the images. For each website, they indicated which version they preferred and which version they trusted more. Trust was operationalised as a forced choice to capture relative credibility judgments between versions.

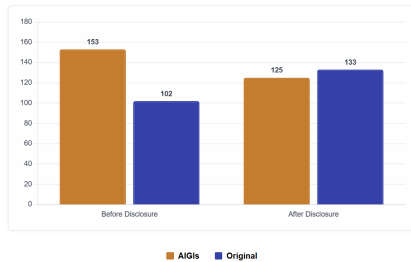
In the second round, the same websites were shown again in a new random order. Participants were informed which version used AI-generated imagery and again indicated which version they preferred and which they trusted more. The final questionnaire collected open-ended responses and self-reports on AI familiarity and whether AI-generated imagery had been suspected before disclosure. Figure 2 illustrates the workflow used to generate and integrate the AI-generated images into the website stimuli.

## 4 Results

Analyses focused on within-participant comparisons between website versions before and after disclosure. Paired statistical tests appropriate for categorical data were used to examine changes in preference and trust across website contexts.

### 4.1 Preference

Before disclosure, participants more often preferred the AIGI versions, accounting for 60% of valid preference choices (153/255). After disclosure, preferences moved toward parity, with slightly more choices favoring original-image versions (133/258) than AIGI versions (125/258; Fig. 3).

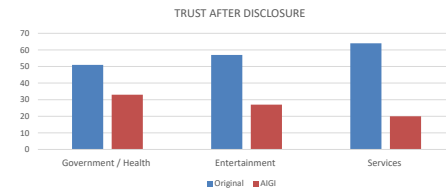


**Figure 3: User preference for websites with AI-generated versus original images before and after AI disclosure.**

A McNemar test indicated a significant shift in preferences after disclosure, with more participants switching from preferring AIGI versions to preferring original-image versions than the reverse ( $p < .001$ ). Preference patterns differed across websites both before disclosure (Cochran's  $Q(5) = 20.09$ ,  $p = .001$ ) and after disclosure ( $Q(5) = 14.35$ ,  $p = .015$ ), suggesting that acceptance of AIGIs depended on the specific website context rather than image origin alone.

### 4.2 Trust

Before disclosure, trust judgments were comparatively balanced between AIGI and original-image versions (134 vs. 121 out of 255 choices). After disclosure, trust shifted strongly toward the original-image versions, which were chosen in 175 of 258 trust judgments (Fig. 4). A McNemar test confirmed a significant post-disclosure shift in trust ( $p = .001$ ), with substantially more participants switching trust from AIGI versions to original-image versions than in the opposite direction. Trust judgments did not significantly differ across websites within each round (Cochran's  $Q$  before disclosure:  $p = .123$ ; after disclosure:  $p = .119$ ). However, the disclosure effect was consistent across categories, with original-image versions trusted more in all cases after disclosure.



**Figure 4: Trust judgments after AI disclosure, aggregated by website category. Across all categories, websites using original images were trusted more than those using AI-generated images, with the largest differences observed for government/health and service websites.**

### 4.3 AI Suspicion and Trust

At the end of the survey, participants reported whether they had suspected AI-generated imagery prior to disclosure. Suspicion was associated with lower trust in AIGI versions during the no-disclosure round ( $\chi^2(1, N = 258) = 8.43$ ,  $p = .004$ ). Participants who suspected AI earlier were less likely to trust AIGI versions even before explicit disclosure, suggesting that individual differences in AI awareness or scepticism shaped trust judgments independently of labels.

### 4.4 Preference–Trust Dissociation

Preference and trust did not consistently align. McNemar tests comparing preference and trust choices revealed a mismatch both before disclosure ( $p = .017$ ) and more strongly after disclosure ( $p < .001$ ). Participants often preferred the visual appearance of AIGI versions while placing greater trust in the original-image versions, highlighting a divergence between aesthetic appeal and perceived credibility.

## 5 Discussion

Our findings show that the effects of AI-generated images in website contexts depend less on visual quality alone and more on disclosure, domain expectations, and users' interpretations of authenticity. Across all websites, disclosure consistently shifted trust toward versions using photographic imagery, even when AI-generated images were initially preferred. This extends prior work on disclosure

effects in advertising and other settings [7, 17] to organisational websites, where credibility and legitimacy are central concerns.

Importantly, responses to AI-generated imagery were not uniform across contexts. Preference patterns differed across websites, and trust penalties after disclosure were strongest for government and healthcare sites, while entertainment websites showed greater tolerance. This suggests that users apply different authenticity standards depending on a website's purpose and perceived stakes. In domains associated with public responsibility or care, AI-generated imagery may be seen as less appropriate or less trustworthy, whereas entertainment contexts allow greater latitude for synthetic or stylised visuals. These findings align with prior work showing that website evaluations vary by category and purpose [16], and extend this insight to AI-generated imagery.

Open-ended responses further highlight a recurring tension between aesthetic appeal and perceived authenticity. Participants often described AI-generated images as polished and professional, supporting initial preferences, yet associated disclosure with artificiality, manipulation, or a lack of genuineness. This tension helps explain the observed dissociation between preference and trust: participants could favour the appearance of AI-generated imagery while simultaneously trusting original imagery more. Together, these findings caution against assumptions that increasing technical realism or visual quality alone will ensure acceptance in credibility-sensitive contexts.

From a design perspective, our results suggest that the use of AI-generated imagery in organisational websites should be approached with care. In domains where trust and authenticity are critical, AI-generated images may enhance first impressions but pose risks once their origin is revealed. Designers might consider hybrid strategies, such as reserving photographic imagery for representations of people, places, or critical real-world claims, while using AI-generated visuals for abstract or decorative elements. In entertainment contexts, AI-generated imagery appears more acceptable, although disclosure can still have a negative impact on trust.

## 6 Limitations and Future Work

This study has several limitations. The participant sample was recruited through convenience sampling and consisted primarily of young adults, which limits generalisability. We examined a small set of websites and broad categories; future work should include a wider range of domains and multiple sites per category. In addition, the rapid evolution of generative image models and growing public familiarity with AI may alter perceptions over time. Finally, our study does not isolate which specific visual features drive trust penalties. Future research using controlled manipulations and mixed-effects models could help clarify how factors such as subject matter, level of polish, or perceived realism contribute to trust judgments.

## 7 Conclusion

In a within-subjects website experiment, AI-generated images increased preference before disclosure, while disclosure consistently shifted trust toward websites using photographic imagery. These effects were category-sensitive, with greater tolerance for AI-generated imagery in entertainment contexts than in government and healthcare settings. Taken together, our findings indicate that acceptance

of AI-generated images on websites is shaped by domain-specific authenticity expectations and that disclosure reshapes the relationship between preference and trust. This creates a central design tension between aesthetic appeal and perceived genuineness in credibility-sensitive web contexts.

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A Appendix

Table 1: Overview of websites used in the study.

| Website               | Type       | Accessed      | Link                                                                                |
|-----------------------|------------|---------------|-------------------------------------------------------------------------------------|
| Government of Iceland | Government | March 1, 2025 | <a href="https://www.government.is">https://www.government.is</a>                   |
| NSW Government        | Healthcare | March 1, 2025 | <a href="https://www.health.nsw.gov.au">https://www.health.nsw.gov.au</a>           |
| High Score Arcades    | Arcade     | March 1, 2025 | <a href="https://www.highscorearcades.co.uk">https://www.highscorearcades.co.uk</a> |
| Oceanário de Lisboa   | Aquarium   | March 1, 2025 | <a href="https://oceanario.pt/en/">https://oceanario.pt/en/</a>                     |
| John Reed Fitness     | Gym Centre | March 1, 2025 | <a href="https://johnreed.fitness/en-at">https://johnreed.fitness/en-at</a>         |
| Deutsche Bahn         | Railway    | March 1, 2025 | <a href="https://int.bahn.de/en">https://int.bahn.de/en</a>                         |